two as it passed through the strong gravitational tides of a planet like Jupiter; more interesting is the possibility that two worlds on similar orbits made a gentle overtaking collision and stuck. This process may have been key to the building of planets and the Earth. At least one asteroid (Ida, as viewed by *Galileo)* has its own small moon. We might guess that two asteroids in contact and two asteroids orbiting one another have related origins.

Sometimes, we hear about an asteroid making a "near miss." (Why do we call it a "near miss"? A "near hit" is what we really mean.) But then we read a little more carefully, and it turns out that its closest approach to the Earth was several hundreds of thousands or millions of kilometers. That doesn't count—that's too far away, farther even than the Moon. If we had an inventory of all the near-Earth asteroids, including those considerably smaller than a kilometer across, we could project their orbits into the future and predict which ones are potentially dangerous. There are an estimated 2,000 of them bigger than a kilometer across, of which we have actually observed only a few percent. There are maybe 200,000 bigger than 100 meters in diameter.

The near-Earth asteroids have evocative mythological names: Orpheus, Hathor, Icarus, Adonis, Apollo, Cerberus, Khufu, Amor, Tantalus, Aten, Midas, Ra-Shalom, Phaethon, Toutatis, Quetzalcoatl. There are a few of special exploratory potential—for example, Nereus. In general, it's much easier to get onto and off of near-Earth asteroids than the Moon. Nereus, a tiny world about a kilometer across, is one of the easiest.* It would be real exploration of a truly new world.

Some humans (all from the former Soviet Union) have already been in space for periods longer than the entire round-trip time to Nereus. The rocket technology to get there already exists. It's a much smaller step than going to Mars or even, in several respects, than returning to the Moon. If something went wrong, though, we would be unable to run home to safety in only a few days. In this respect, its level of difficulty lies somewhere between a voyage to Mars and one to the Moon.